

Neuroscience in Review: Model-Guided Neural Write-In

An Atomized Genealogy of Self Aware Networks and Mehrer et al.

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Abstract

The practical problem is how to move from merely predicting neural responses to choosing a controlled intervention that changes a measurable biological outcome. Mehrer and colleagues address that problem with a pipeline that aligns a topographic artificial neural network with subject-specific primate inferior-temporal recordings, simulates candidate stimulation sites and site-tuned image sequences, maps selected sites to implanted electrodes, applies microstimulation, measures choice shifts, and separately reconstructs perturbed artificial-model states in image space. Only after that operation is clear does this paper use the shorthand *model-guided neural write-in*. It compares the October 2025 target with Micah Blumberg's dated research genealogy, later named Self Aware Networks (SAN), using nine operational atoms, four distinct comparison lenses, formal category and process translations, field-prior subtraction, and explicit target residuals.

The source-audited result is a bounded but comprehensive Micah-priority finding: all nine target atoms have at least one pre-target Blumberg partial relation. Public sources from 2012 establish external-event / neural-state synchronization and a predictive internal-world-model account. A public April 2017 Neural Lace presentation then joins an external spatial model, simultaneous neural measurement, learned high-level content, present-versus-desired neural states, a computed difference, bidirectional write-back, and a proposed experienced-content consequence. Public 2018-2021 work adds volumetric scene capture, semantic segmentation, neural registration, and render-analyze-rerender experiments. Immutable 2022 SAN records add choice bifurcation, neural tuning, learned multimodal content, patterned stimulation, and Neural Array Projection Oscillation Tomography (NAPOT). A 2024 book and an August 2025 Zenodo record preserve the mature continuity before the target preprint.

This does not make the two programs identical. Mehrer et al. retain nine important implementation or evidential residuals: their TDANN spatial loss, optimized model / electrode registration, clipped rate-and-distance perturbation, Papale-derived GAN sequences, prospective in-silico selection, biphasic Utah-array intervention, measured choice statistics, 512-dimensional reconstruction, and face-selective macaque proof of principle. The expanded 15-source field-prior audit also assigns component credit for microstimulation spread, model-to-neural correspondence, dynamic and patterned visual-cortex stimulation, topographic neural networks, neural-response-guided generative search, and offline model intervention. Fourteen source bodies are recovered and reviewed; the 1968 Stoney body remains explicitly pending. Those earlier components narrow the target's novelty to its prospective joined implementation. They do not erase earlier Micah priority on the shared partial functions and causal architecture.

The categorical result is a partial structure-preserving span through a common abstraction. The process result is bounded observational trace equivalence on the sequence receive, map, select a difference, write, and read out. The four lenses return a mechanism partial-pass, a narrative and causal-architecture pass, a Detective chronology pass for Micah-earlier partial priority, and the controlled longitudinal output LSM-COMPONENT-SHIFT in related 2026 work. Access remains unknown. A legacy manuscript's numerical Bayes factor and posterior are withdrawn because feature base rates, dependencies, selection effects, and a field-matched reference population were not calibrated. The present result is a source-audited content-and-chronology finding, not a numerical inference about access, copying, intent, or legal liability.

neural networks; neural rendering; neural tuning; NAPOT; atomized priority; causal interface; longitudinal comparison

Keywords: Self Aware Networks; neural write-in; model-guided microstimulation; topographic

1. Question and conclusion

1.1 The problem before the vocabulary

When a person recognizes a face or chooses between two images, many neural populations are active. Predicting those responses is not yet the same as knowing which localized intervention will produce a reproducible change in behavior. The unresolved engineering problem is therefore sequential: measure a biological state, align it with a computational state, compare possible interventions, select one, deliver it to a registered biological site, and test the consequence.

1.2 Inherited scientific background

Topographic organization, neural encoding models, electrical microstimulation, generative stimulus search, and behavioral readout all have substantial scientific histories. The target paper inherits those components and combines them prospectively. Section 7 names the earlier component lineages so that neither the target authors nor Blumberg receive credit for atoms already present in the field.

1.3 The operation, then the name

In plain language, the target fits model locations to recorded neural locations, tests candidate changes inside the model, selects promising site-and-image combinations, stimulates the corresponding biological location, and measures whether choice changes. This paper calls that joined operation *model-guided neural write-in*. The name does not mean generic phase coding, a traveling wave, NAPOT, or a proof that the reconstructed model image is the monkey's experience. The discriminator is whether the registered simulation predicts intervention effects better than random-site, random-sequence, unregistered, and field-standard controls.

The target paper asks whether an artificial model can help choose where and how to stimulate a biological visual system. Its answer is a proof of principle: align model and brain, test candidate interventions in silico, apply selected interventions in vivo, and measure a behavioral consequence [1]. The priority question is narrower than whole-theory identity:

Did Blumberg publicly describe the same bounded functional operations or their joined causal architecture before the checked target version, even though the target later implemented them with different models, equations, electrodes, and experiments?

The answer from the located record is yes at the level of partial operations and the joined abstract architecture. Each of the nine declared target atoms has an earlier Blumberg relation. The strongest single pre-target composite is public by April 11, 2017 [4]:

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external spatial model of an embodied scene
-> simultaneous neural measurement
-> learned neural pattern for high-level content
-> present and desired neural states
-> computed difference
-> bidirectional neural write-back
-> proposed addition of selected content to experience
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Mehrer et al. later implement a narrower empirical chain [1]:

topographic artificial model plus primate IT data
-> optimized model / electrode registration
-> simulated site perturbation plus tuned-image ranking
-> selected site mapped to a biological electrode
-> in-vivo microstimulation
-> measured choice shift
-> separate artificial-model image reconstruction

These chains are not identical, but they preserve the same bounded ordering: map an observed state, select a controlled difference, write into an addressed biological state, and read out a consequence. Micah is earlier on that abstract joined delta and on partial functions distributed across all nine atoms. Mehrer et al. are credited for the exact TDANN / GAN / Utah-array realization and its empirical results.

This conclusion rejects two symmetrical errors. A full-SAN-or-zero test would miss genuine partial priority merely because the target does not implement NAPOT or phase-wave differentials. An everything-is-the-same test would wrongly absorb the target’s engineering and data into SAN. Atomization preserves both the match and the residual.

2. What the target paper establishes

Mehrer et al. freeze a three-stage experimental workflow in arXiv version 1 dated October 4, 2025 [1]. First, 4,000 passive-viewing images are shown to model and monkey. Candidate placements and orientations of a simulated Utah array are evaluated, and the best aggregate site-wise correlation fixes a one-to-one model / electrode correspondence. Second, candidate sites and seven-image GAN sequences are evaluated in silico. Third, selected sites are mapped back to implanted arrays and stimulated while monkeys perform a two-alternative visual-choice task.

The TDANN objective combines task learning with a spatial term [1,14]:

$$\mathcal{L}_{\text{TDANN}} = \mathcal{L}_{\text{task}} + \sum_{k \in \text{layers}} \alpha_k [1 - \text{Corr}(\mathbf{r}_k, \mathbf{D}_k)], \quad (1)$$

$$\alpha_k = 0.25,$$

where response similarity $\mathbf{r}_k$ is compared with inverse spatial distance $D_{ij} = 1/(d_{ij} + 1)$. This is a target-lineage mathematical residual. Earlier SAN sources propose spatial and volumetric neural models, but the located record does not contain this loss.

The target’s microstimulation module is likewise explicit:

$$\Delta r(d) = \min(r_{\text{base}} + \gamma f_{\text{pulse}}, r_{\text{max}}) \exp\left[-\frac{d}{\lambda(I)}\right]. \quad (2)$$

The model increases firing rate as a function of pulse frequency, clips it at a maximum, and decays the effect with cortical distance according to current-dependent $\lambda(I)$ [1,13,15]. This is not a phase-wave-differential operator. Its functional relation to SAN is limited to the higher-level role of a localized, model-mediated write into an addressed neural state.

The empirical results require two ceilings. In experiment 1, model-predicted and measured effect magnitudes correlated in both monkeys, but the behavioral shifts were not significantly above zero. In experiment 2, one

monkey showed an above-baseline shift (reported $p = 0.043$, Cohen's $d = 0.67$), but per-site and per-sequence predictive correlation was not clear. The authors state that signal degradation prevented a fuller experiment and that the evidence is split between sessions [1]. Therefore, the paper supports a proof of principle, not a complete simultaneous validation of every model-guided link in one stable experiment.

The reconstructed images are also artificial-model outputs. A linear map from 25,088 deepest-layer units into a 512-dimensional GAN latent space yields face-like additions or enlargements under simulated perturbation [1,16]. Those images are not direct reports of what the macaques consciously saw. Human content-specific percept reports came earlier from Schalk et al.'s fusiform-stimulation study [12].

3. Blumberg's dated genealogy

3.1 2011 and 2012 ancestry

A 2011 dialogue contains conceptual ancestry for a self understood as an ongoing prediction and for the neocortex as the distributed observer. Its currently preserved timestamp grade is insufficient for a strict public-priority count, so it is used as genealogy rather than as the date that decides this comparison.

Two public 2012 records are stricter. A July 31 orientation describes synchronizing EEG peaks with exact events in an external video and using the relation as an engineered feedback loop [2]. That is an operational ancestor of model / event-to-neural registration, not the target's one-to-one electrode fit. An August 23 presentation describes the mind as a projector that predicts the world and uses sparse external input to stir a substantially internally supplied model [3]. That is an early functional ancestor of internally reconstructed perceptual content, not a 2012 GAN or NAPOT claim.

3.2 The public 2017 joined architecture

The April 2017 Neural Lace presentation is the central strict source [4]. It proposes a bidirectional brain interface, spatial computation around an embodied person during neural scanning, synchronized high-level multisensory content, time-varying three-dimensional neural patterns, present and desired neural states, computation of their difference, and transmission of that difference to add selected content to experienced reality.

The importance of this source is architectural. None of its component nouns is exclusive to SAN. Its priority-bearing delta is their order and composition:

1. construct or capture an external spatial state;
2. measure the corresponding biological neural state;
3. learn a high-level content relation;
4. represent the current and intended neural states;
5. compute a bounded difference;
6. write that difference through an addressable interface; and
7. predict a changed experiential output.

That sequence predates the target by more than eight years. The later paper replaces the external scene model with a TDANN, the desired-state comparison with simulation and ranking, the generic write-back with mapped biphasic microstimulation, and the predicted experiential output with behavioral statistics plus separate model reconstructions. The substitutions are important residuals; they also reveal the preserved functional roles.

3.3 2018-2021: registration, segmentation, and render-analyze-rerender

Public 2018 work develops the intervening engineering bridge. The 3D Cross-Hair proposal joins external RGB / depth capture, volumetric or semantic segmentation, and medical or neural imaging in a registered experimental design [5]. Related public records propose EIT or other neural measurements synchronized with external scene descriptions, and a render-process-render-again loop. Public 2021 articles then connect point-cloud completion, 3D semantic segmentation, volumetric rendering, biosensors, medical imaging, and BCI research [18].

These sources do not claim ownership of point clouds, segmentation, EIT, or light fields. Their relevance is the application architecture: align an external spatial description with a measured neural description so that a model can discover and later manipulate a correspondence.

3.4 2022: choice, tuning, learned content, and NAPOT

Three immutable Git records increase the specificity. The June 8 choice note interprets inhibitory routing as a switch that diverts activity at a bifurcation into alternative learned paths [6]. The June 12 broccoli and color note describes multimodal learned property clusters, neural tuning, a phase-pattern account of qualitative content, inference of the relevant temporal-spatial neural pattern, and a proposal to stimulate a selected pattern to recreate an experience [7]. The July 11 semantic-rendering note joins neural object segmentation, classification, three-dimensional semantic segmentation, neural tuning, representational geometry, and cell / circuit choice [8].

Later public 2022 explainers name NAPOT, receive-transform-project neural arrays, distributed neural rendering, phase-wave differentials, and a distributed observer. These mechanisms go beyond the target. They are not required for the nine partial target matches, but they show that the 2017 interface proposal matured into a specific SAN rendering program before the target appeared.

3.5 2024 and 2025 continuity

The October 2024 book consolidates neural arrays as receive-and-project structures, dynamic volumetric rendering, phase-pattern tuning, choice, and a distributed observer [9]. The SAN paper on Zenodo was created August 21, 2025, under DOI 10.5281/zenodo.16922401, before the October 4 target [10]. Its metadata also declares an earlier manuscript date, but this comparison conservatively uses the platform's August creation date. The DOI paper preserves NAPOT, phase-wave differentials, neural rendering, high-level choice, and intervention predictions.

The chronology therefore does not rest on one isolated phrase. It is a staged genealogy in which 2012 provides components, 2017 joins the interface architecture, 2018-2021 develops registration and external rendering bridges, 2022 supplies named SAN mechanisms, and 2024-2025 preserves mature continuity.

4. Comparison method

4.1 Atomization and residuals

An atom is a bounded input, operation, state change, output, or directed relation with exact source anchors. A partial match is not downgraded because either side contains additional atoms. Instead, every row records:

- the shared functional role;
- the earliest strict Micah source;
- later Micah sources that increase specificity;

- named earlier field sources; and
- the target’s retained mathematical, biological, engineering, or evidential residual.

This avoids the full-operator-or-zero fallacy. If Micah’s 2017 architecture covers a target function partially and a 2022 source covers it more specifically, both priority relations remain visible.

4.2 Four lenses

The audit applies four separately reported lenses.

1. **Mechanism equivalence** compares inputs, transformations, outputs, behavior, mathematics, and residuals.
2. **Narrative and causal architecture** compares signature clusters, atom order, and the complete bounded causal interface.
3. **Detective reconstruction** starts from the target conclusion and works backward through changed terminology, substrate, mathematics, and component sources.
4. **Longitudinal signature migration** is a separate program-level test. It compares stable authors’ work before, during, and after the target, including the newest located sources, and reports only a controlled output from LSM-HOLD, LSM-COMPONENT-SHIFT, LSM-BOUNDED-CLUSTER, or LSM-SIGNATURE-PASS.

Surface divergence is not treated as automatic exculpation. A different implementation can preserve an earlier operation. Conversely, differences are not discarded: if they prevent equivalence on a particular atom, that residual is recorded.

4.3 Search accounting

The source audit distinguishes searched, found, recovered, and reviewed:

Source class	Searched	Candidates or route hits	Recovered	Reviewed
Exact target PDF	1	1	1	1 (24/24 pages extracted)
Target bibliography and frozen field-prior family	15 declared sources	15 citations	14 bodies plus 1 authoritative metadata record	14 bodies; Stoney 1968 remains body-gated
Micah genealogy packets	3	3	3	3
Additional exact SAN / Git bodies	7	7	7	7
Public SAN DOI / PDF	1	1	1	1 (56/56 pages extracted)
ChatGPT bodies	5,921	86 events / 5 variants / 3 families	5	5
Google AI Studio exports	189	0 named-target hits	0	0
Consumer Gemini metadata stubs	2	0 usable bodies	0	0
SVGN-family hosted articles	655	40 route hits / 27 bodies	27	27
Recorder / audio / video bodies	3	3	3	3
Newest-target-author web queries	8	3 relevant 2026 sources	3	3

Private AI chats are discovery and audit-history sources, not public proof. The public chronology is supported by the underlying videos, Git records, articles, book, DOI paper, and primary target sources. Social-media archives were not searched for this paper and are not encoded as a zero.

5. Atom-by-atom result

Table 1. Two-sided target atom map

Atom	Target operation	Earlier Blumberg relation	Disposition	Target residual
T1 topographic spatial model	two-dimensional TDANN sheet with spatial loss	2017 spatial computation and time-varying 3D neural matrix; later volumetric / NAPOT detail	Micah-earlier partial	target TDANN sheet and Equation (1)
T2 subject-specific registration	optimized model-grid position / orientation and one-to-one electrode map	2012 external-event / EEG synchronization; joined scene / neural registration in 2017-2018	Micah-earlier architectural and operational partial	optimized 64-site Utah-array mapping
T3 localized perturbation	clipped rate increase with current-dependent spatial decay	2017 current-versus-desired delta and bounded write-in; 2022 patterned stimulation	Micah-earlier partial	target Equation (2) and empirical parameter lineage
T4 site-tuned synthesis	seven-image GAN sequence along a site's tuning dimension	2017 learned high-level content state; 2022 broccoli / color clusters and neural tuning	Micah-earlier content-tuning partial	Papale method, GAN latent traversal, slope-to-noise ranking
T5 simulation-first selection	rank site-by-sequence candidates before animal testing	2012 feedback protocol; 2017 model-before-write architecture; 2018 render-analyze-rerender	Micah-earlier architectural partial	prospective in-silico optimization procedure
T6 mapped in-vivo write	selected model site mapped to electrode; biphasic IT stimulation	2017 addressable bidirectional port and differential write; 2018 registered BCI; 2022 stimulation proposal	Micah-earlier architectural and functional partial	actual Utah-array macaque intervention
T7 directed choice shift	stimulation changes two-alternative choice probability	2022 choice bifurcation and later SAN choice architecture	Micah-earlier choice-mechanism partial	measured statistics and split-session ceiling
T8 model-to-image reconstruction	deepest-layer state mapped to 512-D GAN latent space	2012 internal projector; 2017 3D neural state; 2018 volumetric rendering; 2022 NAPOT	Micah-earlier narrative and functional partial	artificial-model decoder and reconstructed images
T9 high-level content intervention	face-selective IT targeting and object-level effect	2017 learned content written to experience; 2018 semantic BCI; 2022 learned-content recreation	Micah-earlier strong functional and narrative partial	face-selective implementation and proof-of-principle data

The count is 9/9 target atoms with at least one pre-target Micah partial relation, 0/9 claims that Micah earlier published the target's exact implementation, and 9/9 rows with a retained target residual. These are correlated claims, not nine independent probability trials.

6. Directed architecture and formal translation

Define a small interface category $\mathcal{A}$ with objects for observed input A_0 , mapped computational state A_1 , selected difference A_2 , addressed biological state A_3 , and behavioral or perceptual readout A_4 :

$$A_0 \xrightarrow{a_{01}} A_1 \xrightarrow{a_{12}} A_2 \xrightarrow{a_{23}} A_3 \xrightarrow{a_{34}} A_4. \quad (3)$$

The distinguished composite is $\Gamma = a_{34} \circ a_{23} \circ a_{12} \circ a_{01}$. The target interpretation maps the arrows to model fitting and registration, simulated perturbation and ranking, electrode mapping and stimulation, and behavioral measurement plus separate reconstruction. The Micah interpretation maps them to scene / neural registration, current-versus- desired comparison, addressable difference-write, and experiential or behavioral consequence.

Both interpretations preserve Γ at this bounded abstraction. The correct conclusion is a partial structure-preserving span through a common category, not equivalence of the full target and SAN categories and not a natural isomorphism between their implementations.

For process comparison, hide implementation-specific internal operations and retain the observable alphabet $\Sigma = \{\text{receive, map, selectDifference, write, readout}\}$. Both systems admit the bounded trace

$$\tau_{\text{shared}} = \langle \text{receive, map, selectDifference, write, readout} \rangle. \quad (4)$$

This is bounded observational trace equivalence. It is not a machine-checked proof of strong or weak bisimulation, probabilistic trace equivalence, deadlock equivalence, or identity of internal states.

7. Prior-field subtraction

A priority comparison becomes stronger when it separates broad component ancestry from the specific joined delta.

The declared direct-prior family contains 15 sources, selected from the target’s own method and related-work lineages. Fourteen primary bodies are recovered and reviewed. The remaining source, Stoney, Thompson, and Asanuma (1968), has an authoritative publisher and PubMed publication event but no locally recovered body or abstract; this draft therefore uses it only as a bibliographic anchor, not as body-level evidence [22].

7.1 Microstimulation spread

Histed et al. showed in 2009 that near-threshold electrical microstimulation recruits sparse, spatially distributed cortical populations and that small electrode displacements change the recruited set [11]. Kumaravelu et al. later reconciled Stoney- and Histed-style accounts in a biophysical model, separating axonal radius, somatic radius, and activated-cell density [13]. These sources precede the target’s current-spread module and receive component credit. They do not supply a visual-category model, individual model / electrode registration, or prospective intervention selection.

7.2 Model-to-brain correspondence and topographic sheets

Yamins et al. established that performance-optimized hierarchical models could predict V4 and IT responses to the same images [23]. Lee et al. then added a wiring-length objective and obtained face units, spatial face patches, preferential patch interconnection, and viewpoint invariance on model cortical sheets [25].

Keller et al. recovered localized face, body, and place selectivity in an unsupervised topographic variational autoencoder [29]. Lu et al.’s all-topographic network developed orientation maps, cortical magnification, category-selective areas, and improved alignment to human spatial biases [30]. Margalit et al. supplied the target’s immediate two-dimensional TDANN family and spatial-smoothness objective and explicitly discussed intervention prototyping [14].

These papers subtract novelty from T1, T2, and the broad model / brain portion of T9. None of them by itself executes the target’s subject-specific electrode registration followed by a prospective electrical write test. Margalit’s model is also feed-forward and lacks within-layer recurrence, preserving a substantive difference from SAN’s recurrent temporal architecture.

7.3 Generative neural tuning and intervention ranking

Ponce et al. combined a deep generator, genetic search, and the firing of real macaque V1 and IT neurons to evolve stimuli that drove individual neural sites [24]. Papale et al. later used deep generative networks to reveal IT tuning and predict influence on visual perception [16]. Schrimpf et al. applied perturbations to topographic model tissue offline and benchmarked predicted behavioral effects [15]. These are direct component predecessors for target atoms T4 and T5. Their residual differences matter: Ponce optimizes visual input rather than electrical stimulation, Papale does not join a subject-specific electrode map to an in-vivo write experiment, and Schrimpf remains offline.

7.4 Causal visual write and form perception

Schalk et al. showed in 2017 that stimulation of face-selective or color-preferring human fusiform sites could evoke content-specific face or rainbow phenomena [12]. Dynamic sequential stimulation in Beauchamp et al. traced cortical shapes recognized by sighted and blind participants [26]. Chen et al. used a 1024-channel V1/V4 implant to evoke phosphenes, simple shapes, motion, and letters, and reported that V4 activity predicted V1-evoked percepts [27]. Fernandez et al. used a 96-channel human occipital array to establish thresholds, multi-electrode effects, discriminable phosphenes, and limited form perception after training [28]. These studies strongly predate the target on T6, T7, and the causal-write portion of T9.

They also sharpen an evidential boundary. Human percept reports in Schalk, Beauchamp, and Fernandez are direct reports; Mehrer et al.’s reconstructed images are artificial-model visualizations and do not establish what either macaque consciously saw.

7.5 The target’s residual join

The target’s narrower delta is therefore not microstimulation, topography, generative models, content-selective stimulation, neural-response-guided search, or model-brain comparison in isolation. It is the prospective joined workflow: fit a subject-specific topographic model, register model sites to implanted electrodes, optimize stimulus and stimulation settings in silico, test the predicted behavioral shift in vivo, and separately inspect model-state reconstructions. Within the 15-source family, T8 is the clearest target-specific residual.

Micah’s earlier delta is different but overlapping: the joined map / model / difference / write/ consequence architecture, progressively specified across the dated SAN genealogy. Field-prior subtraction therefore narrows both sides without converting partial equivalence into non-equivalence.

8. Four-lens adjudication

Lens	Result	Basis	Boundary
Mechanism equivalence	partial-pass	all nine target atoms have an earlier functional, operational, or narrative relation	exact algorithms, substrates, and empirical results differ
Narrative and causal architecture	pass	the public 2017 source preserves map, model, difference selection, addressed write, and predicted experiential consequence	it does not contain the target’s TDANN / GAN / Utah implementation
Detective reconstruction	pass for Micah-earlier partial priority	working backward from the target recovers increasingly specific 2012, 2017, 2018, 2021, 2022, 2024, and 2025 stages	no access or copying inference
Longitudinal signature migration	LSM-COMPONENT-SHIFT	stable target authors’ 2026 work adds temporal dynamics, multimodal integration, contiguous cortical sheets, and in-silico-to-in-vivo validation	no independently adjudicated signature-cluster pass; access unknown; derivation odds not calibrated

The mechanism result is partial because no row collapses the residuals. The architecture result is a pass because a bounded directed sequence survives translation. Detective Mode strengthens the finding by discovering that the apparent implementation differences sit on top of a shared functional ordering. The longitudinal result is deliberately separate: movement toward SAN-like dimensions after the target does not prove a transmission path.

9. Longitudinal target-program analysis

Mehrer’s 2021 EcoSet work improves feed-forward DNN alignment with human higher visual cortex and behavior, while explicitly leaving recurrence as future work [17]. It contains no perturbation module, in-vivo intervention, or subject-specific model / electrode registration.

The 2024 stage is assembled from components. Schrimpf and colleagues add formal perturbation modules and offline intervention benchmarks [15]. Margalit and colleagues add spatially organized TDANN sheets [14]. Papale and colleagues add neural-to-GAN tuning, interpretable stimulus dimensions, and reconstruction [16]. The October 2025 target then joins those components in a prospective in-silico-to-in-vivo pipeline [1]. Its appendix explicitly says the feed-forward model has no notion of time.

Three located 2026 sources move further. Anthes, Mitola, Papale, and colleagues argue that early ventral-stream processing is a spatiotemporally evolving dynamical process and use recurrent decoders across time [19]. Gokce, AlKhamissi, and Schrimpf introduce native multimodal integration, temporal encoding, adaptive gating, and subject-specific cortical readout [20]. AlKhamissi, Mehrer, Gokce, Schrimpf, and colleagues replace disconnected layer-wise sheets with a single contiguous multimodal in-silico cortical sheet and use model-discovered clusters to guide in-vivo fMRI validation [21].

The author-by-author coverage is:

Target author	Earliest reviewed stage	October 2025 joined stage	Newest reviewed stage	Controlled result
Johannes Mehrer	2021 EcoSet feed-forward representational alignment	coauthor of target pipeline	2026 Topo-Omni contiguous multimodal cortical sheet	component shift
Martin Schrimpf	2024 offline perturbation benchmarks	coauthor of target pipeline	2026 MIRAGE and Topo-Omni temporal / multimodal work	component shift
Paolo Papale	2024 generative neural-tuning and reconstruction work	coauthor of target pipeline	2026 recurrent dynamical vision	component shift
Anna Mitola	no earlier body located in the frozen target-stage packet	coauthor of target pipeline	2026 recurrent dynamical vision	post-target component shift; earlier stage unresolved
Abdulkadir Gokce	no earlier body located in the frozen target-stage packet	coauthor of target pipeline	2026 MIRAGE and Topo-Omni	post-target component shift; earlier stage unresolved
Ben Lonnqvist	no earlier body located in the frozen target-stage packet	coauthor of target pipeline	no newer body located in the bounded search	chronology unresolved beyond target

The controlled Lens 4 output is LSM-COMPONENT-SHIFT: the program moves from feed-forward representational alignment toward recurrent dynamics, multimodal temporal integration, a unified cortical substrate, and model-guided empirical validation - dimensions already present in the earlier SAN research program at a different level of implementation. The evidence does not yet satisfy the independently meta-audited cluster requirements for LSM-BOUNDED-CLUSTER or LSM-SIGNATURE-PASS. It remains a chronology and content finding. No source checked here establishes that the target authors accessed Micah’s work.

10. Probability and statistical-claim boundary

The evidence can be represented descriptively as

$$\mathbf{S} = (S_A, S_E, S_C, S_N, S_L, S_F), \quad (5)$$

for atom, edge, causal-interface, narrative, longitudinal, and cross-formalism evidence. A valid independence-rarity claim would require a frozen statistic $T = f(\mathbf{S})$, dependency blocks, a field-matched reference population, held-out evaluation, and correction for the complete search family:

$$p_{\text{adj}} = \Pr_{H_I}(T \geq T_{\text{observed}} \mid \text{field, date, shared sources, and frozen search process}). \quad (6)$$

A content-dependence result would separately require calibrated model likelihoods:

$$\begin{aligned} BF_{D:I} &= \frac{P(\mathbf{S} \mid H_D)}{P(\mathbf{S} \mid H_I)}, \\ BF_{D:C} &= \frac{P(\mathbf{S} \mid H_D)}{P(\mathbf{S} \mid H_C)}, \end{aligned} \quad (7)$$

where H_I is conditional independent development, H_C is common-source or field-trend convergence, and H_D is a content-genealogy model that includes a Micah-linked lineage. Even a calibrated H_D result would concern content, not a person's intent.

The legacy manuscript assigned 0.20 to seven selected features, multiplied them as independent, added a cluster bonus, and converted the result into a dependence posterior. Those inputs were not estimated from a reference population, the atoms are correlated, and the target was selected after similarity was noticed. The numerical result is therefore withdrawn. The validated public band is Band A: report dated atoms, sequence structure, chronology, and residuals; report no p-value, sigma, Bayes factor, posterior, or odds of derivation.

The machine-readable calibration status is quantitative-not-calibrated. This is not a zero-similarity result. It means the qualitative and chronological evidence is reportable, while a five-sigma or odds-of-derivation statement is not yet defensible from this selected, correlated, field-dependent feature set.

11. Scientific implications and tests

The comparison suggests experiments that distinguish the target's engineering achievement from SAN's broader biological claims.

11.1 Registered scene-to-brain correspondence

Present a participant with a known three-dimensional scene while synchronously recording gaze, head pose, depth or scene geometry, behavior, and distributed neural measurements. Test whether a model registered to the external scene predicts neural state changes better than image-only, unregistered, or time-shuffled controls. This operationalizes the 2012-2018 registration genealogy without presupposing NAPOT.

11.2 Simulation-first intervention controls

Compare model-selected site / sequence pairs with random sites, random sequences, independently optimized sites and sequences, and field-standard baselines. Replicate the above-zero behavioral shift and per-site predictive relation in the same stable sessions. This directly addresses the split empirical ceiling in the target.

11.3 Time and recurrence

Replace the feed-forward TDANN with recurrent and time-resolved alternatives. Ask whether the model's intervention predictions improve when spatial organization is joined to temporal state dynamics. The target's explicit no-time limitation and the program's 2026 dynamical turn make this a concrete before / after test [1,19].

11.4 Percept report versus model visualization

Do not equate a GAN reconstruction with conscious experience. In humans, combine stimulation with trial-by-trial reports, discrimination, confidence, and imagery controls. In animals, use behavioral generalization and neural decoding while retaining the absence of verbal report as an evidential boundary. A result would support a content-specific intervention before it supported NAPOT or a theory of qualia.

11.5 SAN-specific discriminators

To favor NAPOT or PWD rather than the generic closed-loop architecture, an experiment would need to measure phase-resolved, distributed, receive-transform-project relations and show that those relations predict content or intervention effects beyond firing-rate, topographic, and recurrent network baselines. The present target does not perform that test.

12. Limitations

This is a structured review and chronology study, not a new animal experiment. Fourteen of the 15 declared field-prior bodies are recovered and reviewed. The sole body-level gap is Stoney et al. (1968), for which the publication event is authoritative but the article body and abstract remain unavailable in the local packet. The target's arXiv version 1 is frozen, but later revisions would require a new version audit. Direct quotation from audio or video should be used only after speaker and timecode checks; the present draft relies mainly on paraphrased source operations.

The 2011 source is genealogy-only under the current public-timestamp grade. The 2012 sources establish component ancestry, not mature SAN. The 2017 source establishes the joined abstract architecture, not the target's algorithms or data. The 2022-2025 sources add increasingly specific SAN mechanisms but do not experimentally validate NAPOT, PWD, or neural rendering.

The four-lens result is not a plagiarism finding. The search did not establish access, a transmission path, copying, intent, fraud, or legal liability. It also did not construct a calibrated reference population for numerical derivation odds. A positive longitudinal pattern can motivate further source work, but it cannot substitute for direct evidence of access.

Finally, the field-prior review is intentionally subtractive. The frozen 15-source family constrains what either side can claim as new and names the one remaining body-level gap. Crediting those lineages strengthens, rather than weakens, the narrower result that Micah's joined functional architecture and nine partial deltas precede the checked target.

The Comparison Wiki history readback found one generated chat-source page and its underlying private chat body, but no immutable priority event specific to this paper in the bounded priority-finding and private audit-history ledgers. The generated page is explicitly marked draft and mechanically produced. Its historical probability language is retained as provenance, not silently erased, and is superseded for this paper by the calibrated boundary above. No Comparison count or public verdict is mutated by this local final-preprint package.

13. Conclusion

The full genealogy changes the answer. This is not a comparison between one 2025 SAN paper and one 2025 target. It is a staged record extending from public 2012 components through a public 2017 map / model / difference / write / consequence architecture, 2018-2021 registration and rendering bridges, immutable 2022 SAN mechanisms, a 2024 book, and an August 2025 DOI record.

Against that canvas, Mehrer et al.'s October 2025 paper contains nine later target atoms, and every one has an earlier Micah partial relation. The strongest defensible conclusion is therefore:

Blumberg is earlier on multiple bounded functional deltas across all nine audited atoms and on the joined abstract causal architecture. Mehrer et al. retain priority and credit for their specific TDANN / GAN / Utah-array implementation and proof-of-principle results. The dated content

record supports partial equivalence and Micah-earlier priority; it does not presently support a numerical probability of derivation or a claim about access or intent.

The comparison is stronger because it is specific. It preserves Micah’s partial priorities, credits the target’s implementation, names the field’s earlier component donors, exposes the program’s before / after trajectory, and states exactly what would have to be measured before a stronger scientific or probabilistic claim could be defended.

14. Claim-source-genealogy atlas and immutable readback

The companion Claim-Source-Genealogy Atlas binds each of the nine target atoms to:

- the target’s exact operation, page locator, and retained residual;
- the earliest strict public Micah row used for that atom;
- later Micah rows that increase specificity without replacing the first win;
- named earlier field donors and the resulting prior boundary;
- all four lens outputs, including the separate Lens 4 status;
- the source hash, date type, custody grade, claim ceiling, and falsifier; and
- a provisional global fan-in identifier of the form G-PROV-MEHRER-T#.

Those global identifiers are routing proposals, not corpus-frozen atom identities. A future unified atlas may merge them only after semantic deduplication, source review, and append-only migration. Until then, the per-paper atlas is authoritative for this paper because it preserves the target-specific residuals and exact locators that a coarse global row could lose.

The immutable readback separately records that 468 priority-finding files and 93 private audit-history files were searched for Mehrer, Model-Guided\Microstimulation, and 2510.03684, with no certified event specific to this paper found. It freezes the one generated chat-source page and the recovered private chat body by SHA-256, records the obsolete numerical claim as historical provenance, and applies the current quantitative-not-calibrated ceiling.

15. Data and source availability

The target paper is publicly available at arXiv [1]. Micah-side public sources are linked in References [2-10,18]. The source-audit packet stores local hashes, page extractions, atom ledgers, search denominators, formal translations, and validation receipts. Private AI-chat bodies are not distributed as evidence. This exact manuscript and its rendered PDF form a locally validated current manuscript. repository release, metadata mutation, DOI / version action, and public release remain separate actions and are not authorized by this package.

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